

ABE Reviewer 2024

Volume 1

***Agricultural and Biosystems
Power Engineering***

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Agricultural and Biosystems Power Engineering

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ABE Licensure Exam Coverage

(Subjects, Topics and Subtopics are as specified in 2024 TOS)

Subject A	AGRICULTURAL AND BIOSYSTEMS POWER, ENERGY AND MACHINERY ENGINEERING AND ALLIED SUBJECTS (32%)
Topic I	Agricultural and Biosystems (AB) Power Engineering (4%)
Topic II	AB Mechanization Planning, Operation, Maintenance, Management and Manufacturing (8%)
Topic III	AB Machinery Specifications, Testing and Evaluation (3%)
Topic IV	AB Automation, Instrumentation and Control System (4%)
Topic V	Project Management, Feasibility Study Preparation/Evaluation, Research, Development and Extension, and Information System on AB Engineering (7%)
Topic VI	Laws, Professional Standards and Ethics (6%)
Subject B	LAND AND WATER RESOURCES ENGINEERING AND ALLIED SUBJECTS (32%)
Topic I	Hydrology (2%)
Topic II	Irrigation and Drainage Engineering (8%)
Topic III	Soil and Water Conservation Engineering (6%)
Topic IV	Aquaculture Engineering (2%)
Topic V	Fundamentals of Agricultural, Fishery, Ecological and Environmental Sciences (6%)
Topic VI	Mathematics and Basic Engineering (8%)
Subject C	AGRICULTURAL AND BIOSYSTEMS STRUCTURES, ENVIRONMENT ENGINEERING, BIOPROCESS ENGINEERING AND ALLIED SUBJECTS (36%)
Topic I	Agricultural Buildings and Structures (9%)
Topic II	Farm Electrification (4%)
Topic III	Environment Engineering (5%)
Topic IV	Agricultural and Bioprocess Engineering (14%)
Topic V	Food Engineering (4%)

Categories, their Descriptions & Percentages in the Exam

(Easy - 30%, Moderate - 50%, Difficult – 20%)

Easy1	Remembering. Recalling information, recognizing, listing, describing, retrieving, naming, finding
Easy2	Understanding. Explain, interpret, summarize, examine, paraphrase, classify
Moderate	Applying. Using the information in another familiar situation, implementing, carrying out, using, executing
Difficult1	Analyzing. Understanding, comparing, organizing, deconstructing, interrogating, finding
Difficult2	Evaluating. Hypothesizing, critiquing, experimenting, judging
Difficult3	Creating. Generating, designing, constructing, planning, producing, inventing

Internal Combustion Engines – Parts & Operation

Instructions:

1. Select the best answer.
2. Shade only one answer per question in the answer sheet.
3. Don't make unnecessary marking in your answer sheet.

Questions:

1. In a 4-stroke diesel engine, which stroke produces power to do mechanical work?

A. Induction stroke
B. Compression stroke
C. Power stroke
D. Exhaust stroke

Answer: C

2. Based on diesel fuel thermodynamics, high temperature during fuel explosion in an engine results to _____.

A. High combustion chamber pressure
B. Faster rotation of the crankshaft
C. Exhaust valve opening
D. Inlet valve closure

Answer: A

3. Due to small engine space in mobile fisheries equipment, they are best manufactured using a _____ engine.

A. 1-stroke
B. 2-stroke
C. 4-stroke
D. 6-stroke

Answer: B

4. In an internal combustion engine, the _____ converts linear motion of the piston to rotational motion.

- A. Crankshaft
- B. Cam
- C. Connecting rod
- D. Crank

Answer: A

5. In a diesel engine, at what stroke is fuel injected into the cylinder?

- A. Induction stroke
- B. Compression stroke
- C. Power stroke
- D. Exhaust stroke

Answer: B

6. In a diesel engine, the temperature of the compressed air reaches about _____ degrees Fahrenheit.

- A. 130
- B. 1,300
- C. 13,000
- D. 3,100

Answer: B

7. In a gasoline engine, which of the following parts is absent?

- A. Spark plug
- B. Oil ring
- C. Compression ring
- D. Fuel injector

Answer: D

8. In a diesel engine, what causes the fuel to explode?

- A. High compression

- B. High temperature
- C. Friction
- D. Spark

Answer: B

9. Inside the diesel cycle engine, the high temperature needed for fuel explosion is attained at _____.

- A. Fast piston stroke
- B. Ring friction with the cylinder
- C. High fuel pressure
- D. High air pressure

Answer: D

10. The piston of a 4-stroke engine typically has ____ or more compression rings and at least 1 oil scraper ring.

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

11. In the thermodynamics of fuel inside an internal combustion engine, the constant volume cycle is also known as _____.

- A. Otto cycle
- B. Diesel cycle
- C. Carbon cycle
- D. Motor cycle

Answer: A

12. In today's Otto cycle engine, the fuel is _____.

- A. Gun powder
- B. Gasoline
- C. Diesel
- D. Pure bioethanol

Answer: B

13. The fuel for diesel cycle engine is _____.

- A. Gun powder
- B. Gasoline
- C. Diesel
- D. Bioethanol

Answer: C

14. Four-wheel tractors have normally _____ engines.

- A. 1-stroke
- B. 2-stroke
- C. 4-stroke
- D. 6-stroke

Answer: C

15. Due to high power requirement, four-wheel tractors are normally manufactured using _____ engines.

- A. Diesel cycle
- B. Otto cycle
- C. Carbon cycle
- D. Water cycle

Answer: A

16. A high-power diesel cycle engine used in tractors can be fueled by diesel blended with _____.

- A. Bioethanol
- B. Biodiesel
- C. Methanol
- D. Gasoline

Answer: B

17. A high-power Otto cycle engine used in aquaculture engineering can be fueled by gasoline blended with _____.

- A. Bioethanol
- B. Biodiesel
- C. Kerosene
- D. Diesel

Answer: A

18. In Otto cycle engine, fuel explosion occurs at ____.

- A. Constant volume
- B. Constant pressure
- C. Constant temperature
- D. Constant power

Answer: A

19. In diesel cycle engine, fuel explosion occurs at ____.

- A. Constant volume
- B. Constant pressure
- C. Constant temperature
- D. Constant power

Answer: B

20. The ____ controls the opening and closing of intake and exhaust valves of internal combustion engines.

- A. Connecting rod
- B. Piston ring
- C. Crankshaft
- D. Camshaft

Answer: D

21. During power stroke in an engine, the ____ transmits mechanical power to the crank.

- A. Connecting rod
- B. Piston ring
- C. Crankshaft
- D. Camshaft

Answer: A

22. In a gasoline engine, the _____ initiates the fuel explosion.

- A. Fuel injector
- B. Spark plug
- C. Igniter
- D. Friction

Answer: B

23. In a 2-stroke gasoline engine, the crank makes ____ revolution(s) while the piston makes 2 strokes to complete the cycle of events.

- A. 1
- B. 2
- C. 3
- D. 4

Answer: A

24. When the engine is in operation, the pressure inside a fuel injector must be _____ than the pressure inside the diesel engine combustion chamber.

- A. Higher
- B. Twice lower
- C. Thrice lower
- D. 0.5 times higher

Answer: A

25. During combustion inside the engine chamber, the ____ is pushed downward.

- A. Crank
- B. Crankshaft
- C. Piston
- D. Exhaust valve

Answer: C

Repair and Maintenance of Internal Combustion Engines

26. In engine maintenance, when changing the engine oil, the _____ must also be changed.

- A. Air filter
- B. Oil filter
- C. Fuel strainer
- D. Oil strainer

Answer: B

27. Changing the engine oil is regular every certain number of hours used or certain number of kilometers traveled (for vehicles) recommended by the manufacturer but it is mandatory when the oil color is _____.

- A. Brown
- B. Yellow
- C. Black
- D. Colorless

Answer: C

28. In engine repair, what is the first step?

- A. Inspecting and disassembling the engine block
- B. Removing the engine
- C. Disassembling the cylinder head
- D. Reassembling the engine block

Answer: B

29. A long metal for determining the correctness of oil level during regular inspection of big engines.

- A. Oil stick
- B. Oil gauge
- C. Oil dipstick
- D. Oil meter

Answer: C

30. A valve spring compressor looks like a _____.

- A. Hack saw
- B. Vise grip
- C. Pipe wrench
- D. Adjustable wrench

Answer: A

31. When installing new piston rings, the tiny gaps at the end of rings are turned about _____ degrees apart around the piston.

- A. 30
- B. 60
- C. 90
- D. 180

Answer: D

32. A mechanical device for lifting up and lowering down heavy engines.

- A. Hoist
- B. Chain
- C. Jack
- D. Lever

Answer: A

33. Appropriate equipment or tool for installing the engine cylinder sleeves into the engine block.

- A. Hydraulic press
- B. Jack
- C. Hammer
- D. Jack hammer

Answer: A

34. In installing the crankshaft into the engine block, bolt tightening is gradual and simultaneous for all bolts until the torque is about _____ foot-pounds.

- A. 60
- B. 120
- C. 180
- D. 240

Answer: C

35. To keep a tractor engine running seamlessly and efficiently, its maintenance should be _____.

- A. Done when it starts to knock
- B. Regular as recommended by the manufacturer
- C. Once a year
- D. Monthly

Answer: B

36. In dismantling the engine cylinder head, what tool will you use in compressing the valve springs to remove the valve keepers?

- A. Screwdriver
- B. Mechanical pliers
- C. Valve spring compressor
- D. Valve spring remover

Answer: C

37. Using the correct tractor engine oil is essential for keeping your tractor's engine running smoothly and the oil should meet the oil specifications of the _____.

- A. Philippine National Standards
- B. Oil dealer
- C. Tractor supplier
- D. Tractor manufacturer

Answer: D

38. A 4-stroke diesel engine of a rice processing center overheats frequently. Water drops below the engine. What is the most possible cause of overheating?

- A. Malfunctioning radiator fan
- B. Low coolant level
- C. Broken water pump
- D. Low oil level

Answer: C

39. An Otto cycle engine is used in a locally assembled transport vehicle for a sugarcane plantation. It was frequently observed by the operator that the engine stops when it climbs up a slightly sloping farm-to-market road. Analyze. What repair job will you do to solve the problem?

- A. Replace the engine with diesel engine of the same space requirement but of higher rated power
- B. Reduce carriage volume to reduce load
- C. Tighten the timing belt to minimize slippage
- D. Rebore the engine cylinders to increase power

Answer: A

40. A diesel engine starts normally but produces a knocking sound few minutes after starting. The engine has not been opened for relatively longer time. No mechanic is available in the area to open the engine. Analyze. What is the possible cause and what is your first move to alleviate the problem?

- A. Defective spark plugs, clean spark plugs
- B. Malfunctioning fuel injector, replace fuel injector
- C. Broken compression ring, weld compression ring
- D. High carbon deposit at the combustion chamber, use an additive to wear out carbon deposit

Answer: D

41. A transport vehicle smells gasoline in the hood area when running. What part should you inspect immediately and what you should look for?

- A. Fuel tank, for crack
- B. Accelerator pedal, for adjustment
- C. Fuel hose, for leakage
- D. Air conditioner's evaporator, for crack or leakage

Answer: C

42. A rice transport vehicle uses gasoline engine. The engine does not start when the key is turned on and no starting sound can be heard. What part that is connected to the engine will you inspect first?

- A. Spark plug
- B. Battery
- C. Fuel line
- D. Fuel injector

Answer: B

43. A fishing company is planning to buy a 105-hp diesel engine for the rehabilitation of its fishing boat. The company consults you as an AB engineer to analyze and recommend immediately as to what unit to purchase: a 10-year old Php100,000 second hand functional engine or a brand new Php500,000 engine of the same brand? What will be your recommendation?

- A. Buy the 10-year old engine to reduce engine cost
- B. Buy the new engine to reduce maintenance cost
- C. Buy any of the two engines
- D. Defer recommendation for an in-depth analysis

Answer: B

Estimating Power and Energy Production from Renewable Energy Sources & Generators

44. What is the ultimate source of energy?

- A. Fossil fuel
- B. Geothermal
- C. Solar radiation
- D. Sun

Answer: D

45. A tipping bucket rain gage equipped with transmitter was designed to send rainfall data to a distant agrometeorological station. It is to be installed beside a creek in a remote area. It consumes 11.1 milliwatts. What is the appropriate source of power?

- A. Picohydro
- B. Microhydro
- C. Small hydro
- D. Minihydro

Answer: A

46. A picohydro generator is to be used as power source for an analog pressure transmitter sending real-time data on water depth at dam outlet. The pressure transmitter has a voltage of 15 Volts DC and amperage of 20 milliamperes. What minimum power output and voltage of picohydro generator should you request for purchasing?

- A. 36 watts, 15 Vdc
- B. 365 milliwatts, 15 Vdc
- C. 280 milliwatts, 12 Vdc
- D. 250 milliwatts, 12 Vdc

Answer: B

Solution:

Use a minimum safety factor (SF) of 20% or 0.2

Power output = Voltage x Amperage x (1+SF)

$$= 15 \text{ Vdc} \times 20 \text{ mA} \times (1+0.2)$$

$$= 360 \text{ milliwatts minimum}$$

Voltage = System voltage of the device to be used
= 15 Vdc

Note: Efficiency is not used since the power selection is already in terms of power output, not rated power.

47. Estimate the dependable power output from a windpump having a rotor diameter of 8 meters, average blade twist angle of 45 degrees, and tip speed ratio of 1:1 if the daytime prevailing windspeed in the location is 3 meters/second.

- A. 135 watts
- B. 6.1 kW
- C. 163 watts
- D. 3.1 watts

Answer: A

Solution:

Use windpump Rule of Thumb formula, $P=0.1AV^3$

Note: The complicated Windmill Power Output formula that involves air density, rotor power coefficient, transmission efficiency, frictional loss coefficient, rotor area, windspeed, and volumetric efficiency (if applicable), is impractical to be used for small windmills.

Note that $A = 3.14r^2$

$$\begin{aligned} \text{Power output} &= 0.1 \times (3.14 \times 4^2) \times 3^3 \\ &= 0.1 \times 50.24 \times 27 \\ &= 135.6 \text{ watts} \end{aligned}$$

48. The pumping power requirement of a new large livestock farm adjacent to a river is 102 kW. You are the project consultant. What power source will you recommend?

- A. Minihydro
- B. Microhydro

- C. Windpump
- D. Solar pump

Answer: D

49. Estimate the dependable power output and system efficiency of a microhydro having a gross head of 60 meters, minimum water discharge of 150 liters/second, and penstock water velocity of 12 meters/second.

- A. 52 kW, 60% efficiency
- B. 635 kW, 60% efficiency
- C. 75 kW, 85% efficiency
- D. 6.2 kW, 85% efficiency

Answer: A

Solution:

The typical microhydro efficiency is 60% or 0.6

$P_o = 9.8QHE$, P = Power Output in kW, Q = discharge in m^3/s , H = gross head in m, E = efficiency in ratio

$$\begin{aligned} P_o &= 9.8 \times (150/1,000 \text{ m}^3/s) \times 60 \text{ m} \times 0.6 \\ &= 52.9 \text{ kW} \end{aligned}$$

50. Determine the daily energy output of a 24-volt DC solar pump during a sunny day in Luzon with latitude of 16 degrees north if its overall power output is 750 watt-peak and the daily sensible sunlight in the location is 5 hours.

- A. 30 KWH
- B. 6.3 KWH
- C. 3.75 KWH
- D. 90 KWH

Answer: C

Solution:

The energy output of a solar system is Solar Insolation denoted as I

$$\begin{aligned}
 I &= \text{Power output in kW} \times \text{hours of sensible sunlight} \\
 &= 0.750 \text{ kW-peak} \times 5 \text{ hours} \\
 &= 3.75 \text{ kWh}
 \end{aligned}$$

51. A solar water heater installed in Mindanao at latitude of 8 degrees north is used for pre-heating water prior to entry into industrial-size boilers. The heater has a collection area of 3.6 meters x 5 meters, and efficiency of 9%. Estimate its power output during a sunny noon and if you are tasked to carry out the installation, what is its installation tilt angle and north-south orientation.

- A. 16 kW, 8 degrees facing north
- B. 6.1 kW, 10 degrees facing south
- C. 1.6 kW, 8 degrees facing north
- D. 1.6 kW, 10 degrees facing south

Answer: D

Solution:

$$\begin{aligned}
 P_o &= \text{Solar irradiance} \times \text{Collector area} \times \text{Efficiency} \\
 &= 1 \text{ kW/m}^2 \times (3.6 \text{ m} \times 5 \text{ m}) \times 0.09 \\
 &= 1.62 \text{ kW}
 \end{aligned}$$

Solar module installation tilt angle
 = Latitude of site but minimum of 10° for self cleaning purposes, facing the opposite hemisphere
 = 10 degrees facing south

52. Water needs to be elevated into a farm house 30 meters above an existing 10-meter high waterfalls. What renewable energy system will you install for pumping water?

- A. Windpump
- B. Ram pump
- C. Submersible pump
- D. Solar pump

Answer: B

53. A cassava flour processing plant requiring 3-MW power is located in a remote area where electricity

grid is 50 km away but has a waterfall inside its perimeter fence with dependable 6 cubic meters per second discharge and 100 meters height. In your calculation, what is the appropriate power source?

- A. Minihydro
- B. Small hydro
- C. Microhydro
- D. Picohydro

Answer: A

54. A 24-volt DC generator is to be acquired to support DC electrical loads including motors. The computed maximum amperage at peak load is 10 amperes. Surge current is assumed 25% higher than the computed peak load amperage. Load safety factor of 20% is to be used. If you are to carry out the acquisition, what minimum wattage of the generator should be selected?

- A. 300
- B. 306
- C. 360
- D. 630

Answer: C

Solution:

$$\begin{aligned} P &= \text{Voltage} \times \text{Surge amperage} \times (1 + \text{Safety factor}) \\ &= 24 \text{ v} \times (10 \times 1.25) \times (1.20) \\ &= 360 \text{ watts} \end{aligned}$$

55. A 360-watt power from a 12-volt DC generator is to be used for AC farm machine by using an inverter. What is the power output from the inverter if inversion efficiency is 85%?

- A. 300
- B. 306
- C. 360
- D. 630

Answer: B

Solution:

$$\begin{aligned} P_o &= \text{Power input} \times \text{Inversion Efficiency} \\ &= 360 \text{ w} \times 0.85 \\ &= 306 \text{ watts} \end{aligned}$$

56. An AC generator is rated at an apparent power of 100 KVA with a power factor of 0.8, what is the real power of the generator in kW?

- A. 74.6
- B. 80
- C. 107
- D. 109

Answer: B

Solution:

$$\begin{aligned} \text{Real power [kW]} &= \text{Apparent Power [KVA]} \times \text{Power factor} \\ &= 100 \text{ KVA} \times 0.8 \\ &= 80 \text{ kW} \end{aligned}$$

57. An AC generator is rated at an apparent power of 100 KVA with a power factor of 0.8, what is the real power of the generator in hp?

- A. 80
- B. 74.6
- C. 107
- D. 109

Answer: C

Solution:

$$\begin{aligned} P_o &= \text{KVA} \times \text{Power factor} \times \text{Conversion factor} \\ &= 100 \text{ KVA} \times 0.8 \times 1 \text{ hp}/0.746 \text{ kW} \\ &= 107 \text{ hp} \end{aligned}$$

58. An AC generator is rated at a real power of 90 kW with an indicated power factor of 0.82, what is the apparent power of the generator?

- A. 80

- B. 74.6
- C. 107
- D. 109

Answer: D

Solution:

$$\begin{aligned}\text{Apparent power} &= \text{Real power} / \text{Power factor} \\ &= 90 \text{ kW} / 0.82 \\ &= 109.8 \text{ KVA}\end{aligned}$$

59. The generally accepted industry standard for power factor of generators.

- A. 80%
- B. 75%
- C. 90%
- D. 100%

Answer: A

60. In DC generators and DC circuits, the power factor is always _____.

- A. 80%
- B. 75%
- C. 90%
- D. 100%

Answer: D

Determining the most appropriate and efficient physical, chemical and thermodynamic properties of fuels and lubricants

61. The property of the engine oil that determines the lubricant's film strength and efficiency in preventing friction between moving parts.

- A. Viscosity
- B. Alkalinity
- C. Detergency
- D. Load carrying capacity

Answer: A

62. The minimum flash point of oil used in engines necessary to prevent fire.

- A. 220 degrees Celsius
- B. Higher than 220 degrees Celsius
- C. Lower than 220 degrees Celsius
- D. 220 degrees Fahrenheit

Answer: B

63. Classify the tandem of properties of liquid fuels below. Which tandem is not applicable in the Philippines when dealing with operation, repair and maintenance of internal combustion engines?

- A. Viscosity and alkalinity
- B. Viscosity and detergency
- C. Alkalinity and detergency
- D. Cloud point and pour point

Answer: D

64. If you are to carry out engine oiling of newly acquired 10 units diesel four-wheel tractors which are still within the break-in period, which engine oil shall you use?

- A. High viscosity petrol engine oil

- B. Low viscosity petrol engine oil
- C. Low viscosity diesel engine oil
- D. High viscosity diesel engine oil

Answer: C

65. Compare the following liquid fuels in terms of specific heat or the amount of energy required to increase the temperature of unit mass of fuel by 1° Celsius. Which one has the lowest specific heat?

- A. Diesel
- B. Gasoline
- C. Kerosene
- D. Ethanol

Answer: A

66. The specific heat of gasoline is 0.53 kcal/(kg °C). If the current temperature of a 1 kg gasoline is 100°C, find the amount of energy needed to further increase it to 110°C.

- A. 4.8 kcal
- B. 5.3 kcal
- C. 5.2 kcal
- D. 20.5 kcal

Answer: B

Solution:

Approach: Apply the specific heat definition

$$\begin{aligned} \text{Energy needed to increase gasoline temperature} &= 0.53 \text{ kcal/(kg}^\circ\text{C)} \times \text{Temp. change in } ^\circ\text{C} \times \text{kg mass} \\ &= 0.53 \text{ kcal/(kg}^\circ\text{C)} \times (110-100) ^\circ\text{C} \times 1 \text{ kg} \\ &= 5.3 \text{ kcal} \end{aligned}$$

67. The specific heat of diesel is $0.48 \text{ kcal}/(\text{kg } ^\circ\text{C})$. If the current temperature of a 1 kg diesel is 200°C , find the amount of energy needed to further increase it to 210°C .

- A. 4.8 kcal
- B. 5.3 kcal
- C. 5.2 kcal
- D. 20.5 kcal

Answer: A

Solution:

Approach: Apply the specific heat definition

$$\begin{aligned}\text{Energy needed to increase diesel temperature} &= 0.48 \text{ kcal}/(\text{kg } ^\circ\text{C}) \times \text{Temp. change in } ^\circ\text{C} \times \text{kg mass} \\ &= 0.48 \text{ kcal}/(\text{kg } ^\circ\text{C}) \times (210-200) ^\circ\text{C} \times 1 \text{ kg} \\ &= 4.8 \text{ kcal}\end{aligned}$$

68. Based on the specific gravity of liquid fuels, which of the following fuels is lighter than water?

- A. Diesel
- B. Kerosene
- C. Gasoline
- D. Diesel, kerosene and gasoline

Answer: D

69. Based on the density of liquid fuels, which of the following fuels can attain the highest tonnage per thousand liters of storage volume?

- A. Crude oil
- B. Gasoline
- C. Diesel
- D. Biodiesel

Answer: A

Solution:

Shortcut approach: Compare fuels with highest and lowest densities only

Crude oil (highest density):

$$\begin{aligned}\text{Tonnage} &= \text{Density} \times 1,000 \text{ L} \times 1 \text{ tonne}/1,000 \text{ kg} \\ &= 0.9 \text{ kg/L} \times 1,000 \text{ L} \times 1 \text{ tonne}/1,000 \text{ kg} \\ &= 0.9 \text{ tonne}\end{aligned}$$

Gasoline (lowest density):

$$\begin{aligned}\text{Tonnage} &= \text{Density} \times 1,000 \text{ L} \times 1 \text{ tonne}/1,000 \text{ kg} \\ &= 0.75 \text{ kg/L} \times 1,000 \text{ L} \times 1 \text{ tonne}/1,000 \text{ kg} \\ &= 0.75 \text{ tonne}\end{aligned}$$

Therefore, crude oil has the highest tonnage.

70. Based on the density of liquid fuels, which of the following fuels require the biggest storage volume per tonne of fuel?

- A. Crude oil
- B. Gasoline
- C. Diesel
- D. Biodiesel

Answer: B

Solution:

Shortcut approach: Compare fuels with highest and lowest densities only

Crude oil (highest density):

$$\begin{aligned}\text{Vol} &= \text{Density reciprocal} \times 1,000 \text{ L} \times 1 \text{ tonne}/1,000 \text{ kg} \\ &= 1 \text{ L}/0.9 \text{ kg} \times 1,000 \text{ kg/tonne} \\ &= 1,111 \text{ L per tonne}\end{aligned}$$

Gasoline (lowest density):

$$\begin{aligned}\text{Vol} &= \text{Density reciprocal} \times 1,000 \text{ L} \times 1 \text{ tonne}/1,000 \text{ kg} \\ &= 1 \text{ L}/0.75 \text{ kg} \times 1,000 \text{ kg/tonne} \\ &= 1,333 \text{ L per tonne}\end{aligned}$$

Therefore, gasoline requires the highest storage volume per tonne.